

ROBO AI - ITP

45 + T Day Industrial Robotics Training Program Curriculum

This Curriculum outlines a 45 + T day industrial robotics training program, divided into five sections: Mechanical & Mathematics, Industrial Design, Electronics & Electrical, Artificial Intelligence/Machine Learning, and ROS/ROS2. **Each section includes a detailed breakdown** of topics covered, estimated time per topic, and potential adjustments based on audience understanding.

Section 1: MECHANICAL & MATHEMATICS Phase (90 min/day)

Please note: The time allocation and topics covered in this section may need to be adjusted based on the target audience and learning objectives.

Day	Topic	Time (min)	Notes
1	Introduction to Robotics	90	- ABB IRB Case Study, Datasheet, CAD, and use-cases - Intro and history of robots - Elements: Base, Controller, Sensors, Actuators, Grippers
2	Manipulator Physics	90	- Mechanics of manipulators - Links and Joints - Kinematic Pairs - Degrees of Freedom and Gruebler's Criterion
3	Manipulator Physics + Math	90	- Inversions of 4-bar linkages - Inversions of slider crank mechanism - Vectors and Matrices introduction
4	Transformation Matrices	90	- Translation - Rotation Matrices - Transformation Matrices - Numerical Problems
5	DH - Matrices	90	- Understanding frames and coordinate systems - Constructing DH table and transformation matrices

6	Forward Kinematics	90	2D and 3D Forward Kinematics
7	Inverse Kinematics + Jacobian	90	2D Inverse Kinematics 3D Forward Kinematics (repeated) - Jacobian Matrix derivation
8	Doubts + Project + Career	90	<i>Problem Set Project</i> <i>Solutions with calculations</i> <i>Encouragement to solve via programming</i>

Section 2: Industrial Design Phase (90 min/day)

Please note: The time allocation and topics covered in this section may need to be adjusted based on the target audience and learning objectives.

Day	Topic	Time (min)	Notes
1	Introduction to CAD	90	- Introduction to Solidworks software - Understanding different features and tools of software - SolidWorks UI/UX - Basic Sketches
2	Sketches	90	- Basic Sketching - Advanced sketching techniques
3	Basic Features	90	- Boss Extrude, Extrude Cut - Linear, Circular Patterns - Mirror tools - Problem solving - real world examples
4	Advanced Features	90	- Loft and Sweep features - Advanced cuts + Problems
5	Advanced Modelling - Industrial	90	- Surfacing - Weldments - Sheet Metal

6	Assemblies	90	Basic, Advanced, and Mechanical Mates
7	2D Drawings	90	- Understanding Assembly drawing and elements - Understanding Tolerance, BoM, Hole, Weld Tables - Part Drawing
8	ANSYS - Part 1	90	- Introduction - Engineering Materials selection and creation - Modelling 1D simulation of Beam - Modelling Fixed support Arm
9	ANSYS - Part 2	90	- Modelling a revolute joint in an arm - Modelling prismatic joint in an arm - Modelling cylindrical joint in an arm - Thermal exposure - example + modelling
10	Doubts + Project + Career	90	Doubt Session + Project + Career Guidance

Section 3: Electronics & Electrical Phase (90 min/day)

Please note: The time allocation and topics covered in this section may need to be adjusted based on the target audience and learning objectives.

Day	Topic	Time (min)	Description
1	Introduction to Electronics	90 min	- Basics of electricity - Semiconductor Devices
2	Introduction to Microcontrollers	90 min	- Microcontroller Architecture - Bootloaders in Microcontrollers
3	Sensor Integration and Microcontroller	90 min	- Introduction to Wokwi Simulator - Integration of Ultrasonic Sensor with Arduino - Understanding and Implementation of PWM(Pulse Width Modulation)
4	UART and	90 min	Protocol Concept and Waveforms

	I2C Protocol		• Implementation of I2C using MPU6050 Sensor on Wokwi. • Implementation of UART using Arduino on Wokwi
5	SPI Protocol	90 min	• Protocol Concept and Waveforms • Making of a custom SPI chip and library on Wokwi
6	SD Card and LCD Interfacing with Arduino	90 MIN	• Understanding datasheet of LCD • Understanding the grid access of LCD • Implementing non-blocking and Blocking read-write to SD card
7	Buck-Boost Converter	90 min (hands-on)	• Understanding the circuit operation • https://www.falstad.com/circuit/
8	VFD(Variable Frequency Drive)	90 min	• Understanding the need of 3-Phase AC. • Making and understanding Waveforms of the output of the VFD
9	PCB Designing Part 1	90 min	• Overview of the KiCAD software • Importance of Reading datasheets of components. • Making schematic of a circuit • Free CAD Models for PCB: ◦ https://www.ultralibrarian.com/ ◦ https://componentsearchengine.com/ ◦ https://www.snapeda.com/
10	Introduction to IOT	90 min	• Understanding PCB routing rules. • Making a PCB layout based on the schematic made in Part 1
11	Doubt Session + Project + Career Guidance	90 min	• Obstacle Avoiding Robot with Telematics Feature

Section 4: Artificial Intelligence/ML/DL Phase - (90 min/day)

Please note: The time allocation and topics covered in this section may need to be adjusted based on the target audience and learning objectives.

*** Kindly note that the reason for not mentioning the days , is because there can be a requirement for more sessions to complete the mentioned notes below which is subjective to participant's discretion ***

Topic	Time (min)	Notes
Python & Virtual Environment	90	• Introduction to Python: Features, Syntax, and Data Types • Comparison with other languages , Data types • Operators ,conditional statements, Collections ,Loops, string manipulation ,Slicing , Indexing, Formatting , Splitting , Joining , Classes ,objects • Python Environment Setup- Setting up Virtual Environments with venv and virtual env- Installing Python • Packages and Dependencies- Introduction to IDEs • Editors (VSCode, PyCharm)- Writing and Running Basic Python Scripts
Introduction to Machine Learning	90	• What is Machine Learning? • Overview and Types of Learning- Supervised vs Unsupervised Learning- • Introduction to scikit-learn: Installation and Basic Operations- Classification Algorithms: Logistic Regression, k-NN, Decision Trees- • Regression Algorithms: Linear Regression, Polynomial Regression- • Support Vector Machine (SVM): Concept and Mathematical Explanation- • Least Squares Regression: Concept and Implementation- • Reinforcement Learning: Overview and Basic Concepts- • SVM Programming Example (Hands-on)

Introduction to Deep Learning: Artificial Neural Networks (ANN)		• Introduction to Neural Networks and Deep Learning • Understanding a Single Perceptron and its Functioning • Multi-Layer Perceptron (MLP) and its Architecture • Feedforward and Backpropagation Algorithms: Explanation and Applications • Hands-on with MNIST Dataset: Training a Simple Neural Network from Scratch
Introduction to Deep Learning: Convolutional Neural Networks (CNN)	90	• Introduction to Convolutional Neural Networks (CNNs)- CNN Layers: • Convolution, Pooling, Fully Connected Layers • CNN Architecture: Overview and How It Works- Training a CNN Model: Cat vs Dog Classification using CNN • (Hands-on)- Understanding Image Data: Preprocessing, Augmentation, and Transfer Learning
Introduction to Generative AI	90	• What is Generative AI? • Overview and Applications • Generative AI vs Deep Learning: Key Differences • Types of Generative AI Models: Language Models (e.g., GPT, T5) - Image Models (e.g., DALL·E, Stable Diffusion) • Introduction to Autoencoders and Variational Autoencoders • Generative Adversarial Networks (GANs): Concept, Architecture, and Training Process • Hands-on: Training a GAN on MNIST Dataset • Introduction to Hugging Face: Overview and Setup • Accessing Pre-trained Models • Using Transformers and other Hugging Face Libraries • Creating a Chatbot using Hugging Face Models (Hands-on)

Doubt Session + Project + Career Guidance - 90 min

Section 5: ROS/ROS2 Phase - 90 min/day

Please note: The time allocation and topics covered in this section may need to be adjusted based on the target audience and learning objectives.

*** Kindly note that the reason for not mentioning the days , is because there can be a requirement for more sessions to complete the mentioned notes below which is subjective to participant's discretion***

Topic	Time (min)	Notes
ROS2 Environment Setup and Ubuntu Essentials	90	• Introduction to VirtualBox and Installing Ubuntu • Basic Ubuntu Terminal Commands: Navigation, File Operations, Package Management • Installing ROS2 (Humble): Sources, Setup, and Environment Configuration- • ROS2 File Structure and Workspace Overview- ROS2 Command-Line Tools: ros2 run, ros2 topic, ros2 service, ros2 node • ROS2 Nodes, Topics, Services, and Parameters • Conceptual Overview- Essential Tools: colcon, rosdep
Creating ROS2 Workspace and Writing Nodes in Python	90	• Creating and Building a ROS2 Workspace (colcon build) • ROS2 Python Node Structure and Best Practices • Writing a Simple Publisher Node in Python • Writing a Simple Subscriber Node in Python • Running Nodes and Inspecting Communication • Using rqt_graph and ros2 topic echo for Debugging • ROS2 Packages and Dependency Management (package.xml, setup.py, rosdep)
Coordinate Systems, TFs, and Robot Modeling	90	• Intro to rviz2, gazebo, Introduction to Degrees of Freedom (DoF) and Coordinate Frames- • Introduction to TF2 in ROS2: Concepts and Usage- Static vs Dynamic Transforms- • Broadcasting and Listening to TFs • URDF (Unified Robot Description Format) Introduction ○ Structure of URDF: ○ Links and Joints- Types of Joints: Fixed,

		○ Revolute, Continuous, Prismatic ● Building a Basic Robot in URDF ● Visualizing URDF in RViz2 ● Simulating URDF Robot in Gazebo
Sensors, SLAM, and Navigation Stack	90	● Recap of DoF ● Overview of Sensors: Types and Roles in Robotics ● IMU: Working Principles and Use Cases- LiDAR: 2D vs 3D, Ranging, and Scan Patterns ● Camera: Types (RGB, Depth, Stereo) and Working Principles ● LiDAR vs Camera Comparison: Pros, Cons, and Use Cases ● Introduction to SLAM: Concepts and Applications ● SLAM Architecture Overview (Front-end, Back-end, Loop Closure) ● Navigation Stack Architecture: Global Planner, Local Planner, Costmaps ● Introduction to Localization in ROS2 ● AMCL (Adaptive Monte Carlo Localization): Working Principle- Practical: Simulating SLAM and Navigation on a Robot in Gazebo
Computer Vision with OpenCV and ROS Integration	90	● Introduction to OpenCV in Python ● Image Operations: Reading, Displaying, Saving Images- Color Spaces: BGR to Grayscale, HSV ● Basic Image Processing: Thresholding, Masking ● Edge Detection (Canny), Blurring, Morphological Operations ● Contour Detection and Object Shape Analysis ● Opening and Processing Video Streams ● Introduction to cv_bridge in ROS2 ● Writing a Python Node to Publish Camera Image to ROS Topic ● Subscribing and Processing Image Data from Camera using OpenCV
Doubt Session + Project + Career Guidance - 90 min		